Greedy Grid

1. Project Overview

The Grid Learning Engine is a modular and rule-driven game engine designed to support multiple grid-based puzzle games. The engine separates core logic from game configurations using JSON files. This allows developers to add new games without modifying the engine code, improving scalability and reusability.

2. Engine Architecture Diagram

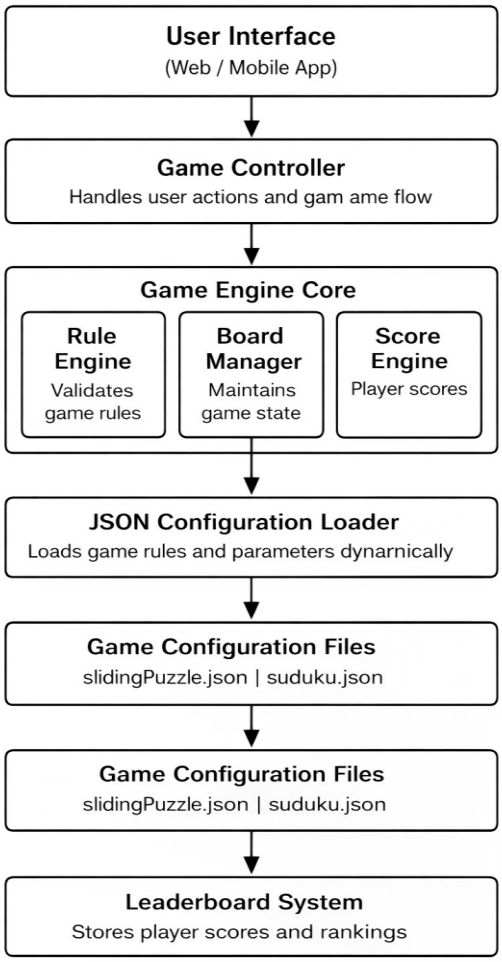

Figure: Engine Architecture Diagram

The UI handles player interaction. The Game Controller processes actions and communicates with the Game Engine Core. The Rule Engine enforces game rules defined in JSON configuration files. The Board Manager maintains the game state, and the Score Engine calculates scores. Game configurations are loaded dynamically from JSON files.

3. JSON Configuration Structure

Each game in the engine is defined using a JSON configuration file. This file specifies board size, difficulty, rules, and scoring parameters. The engine reads these files at runtime and loads the game accordingly.

Example JSON Configuration:

```
{
  "gameId": "sliding-puzzle",
  "gameName": "Sliding Puzzle",
  "boardSize": 4,
  "difficulty": "medium",
  "timeLimit": 300,
  "scoring": {
    "winPoints": 100,
    "movePenalty": 1
  }
}
```

4. JSON Schema Sample

```
{
 "$schema": "http://json-schema.org/draft-07/schema#",
 "type": "object",
 "required": ["gameName", "grid", "rules", "winCondition"],
 "properties": {
 "gameName": {
"type": "string"
}
"grid": {
"type": "object",
"properties": {
"rows": { "type": "integer" },
"columns": { "type": "integer" }
}
},
"rules": {
```

```
"type": "object"
},
"winCondition": {
"type": "object"
},
"scoring": {
"type": "object"
}
} }
```

5. Reusability Design Plan

The engine is designed with modularity in mind. Game logic is separated from configuration. All rules, parameters, and scoring details are defined in JSON files. The core engine remains unchanged when new games are added.

Steps to add a new game:

1. Create a new JSON configuration file.
2. Define the board size, rules, and scoring system.
3. Place the file in the configs/games directory.
4. The engine automatically detects and loads the game.

At runtime, the Game Engine scans the configs/games directory, loads all available JSON configuration files, validates them using the JSON schema, and initializes the corresponding game boards dynamically.

6. Example Project Folder Structure

```
grid-learning-engine/
src/
  engine/
    gameEngine.js
    ruleEngine.js
    scoreEngine.js
    gridEngine.js
configs/
  games/
    slidingPuzzle.json
    sudoku.json
leaderboard/
  leaderboard.json
schema/
  gameSchema.json
docs/
  architecture.pdf
```